

Theorem (Triangle Inequality). *Let $(V, \langle \cdot, \cdot \rangle)$ be an inner-product space and let $\|\cdot\|$ be the norm induced by the inner product, $\|z\| = \sqrt{\langle z, z \rangle}$. For any $x, y \in V$,*

$$\|x + y\| \leq \|x\| + \|y\|.$$

Proof. **1. Degenerate cases.** If $x = 0$ then $\|x + y\| = \|y\|$ and the right-hand side is $\|0\| + \|y\| = \|y\|$; similarly for $y = 0$. Hence the inequality holds with equality. In the sequel we assume $x \neq 0$ and $y \neq 0$.

2. Squaring. Both $\|x + y\|$ and $\|x\| + \|y\|$ are non-negative, therefore

$$\|x + y\| \leq \|x\| + \|y\| \iff \|x + y\|^2 \leq (\|x\| + \|y\|)^2. \quad (1)$$

3. Expansion of the left-hand side. Using linearity in the first argument and conjugate symmetry,

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + \|y\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} \\ &= \|x\|^2 + \|y\|^2 + 2 \operatorname{Re} \langle x, y \rangle. \end{aligned}$$

The right-hand side of (1) expands as

$$(\|x\| + \|y\|)^2 = \|x\|^2 + 2\|x\|\|y\| + \|y\|^2. \quad (3)$$

4. Bounding the mixed term. By the Cauchy–Schwarz inequality,

$$|\langle x, y \rangle| \leq \|x\| \|y\|. \quad (4)$$

Since $\operatorname{Re} z \leq |z|$ for any $z \in \mathbb{C}$, we obtain

$$\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle| \leq \|x\| \|y\|. \quad (5)$$

5. Comparison. Substituting (5) into (1) gives

$$\|x + y\|^2 = \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.$$

Thus (1) holds, and taking square roots yields $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in V$.

6. Equality condition. Equality occurs precisely when equality holds in both (5) and in $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$.

Equality in (5) (Cauchy–Schwarz) means x and y are linearly dependent, i.e. $x = \lambda y$ for some $\lambda \in \mathbb{F}$ (or $y = 0$). Equality in $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$ forces $\langle x, y \rangle$ to be a non-negative real number. Hence $\|x + y\| = \|x\| + \|y\|$ iff either $x = 0$, $y = 0$, or $x = \alpha y$ with $\alpha \geq 0$ (real). In other words, the two vectors point in the same direction (or one is the zero vector). $\square$